Seminar 7: Continuous Random Variables. Normal Distribution

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A discrete random variable assigns positive probability to countably many values.

A **continuous random variable** distributes its probability over intervals rather than individual points. In this lecture we study the most important class of continuous distributions: **absolutely continuous distributions**.

0.1 Probability Distributions

A **probability distribution** on $\mathbb{R}$ is a probability measure

$$P : \mathcal{B}(\mathbb{R}) \to [0, 1],$$

where $\mathcal{B}(\mathbb{R})$ denotes the Borel σ-algebra.

0.2 Cumulative Distribution Function

The cumulative distribution function (CDF) of a random variable ξ is defined by

$$F_\xi(x) = P(\xi \le x), \qquad x \in \mathbb{R}.$$

Notice that the CDF completely determines the probability distribution.

In other words, if two random variables have the same cumulative distribution function, then they have the same distribution.

0.3 Properties of the CDF

1. $F(x)$ is non-decreasing.
2. $\lim\limits_{x\to-\infty} F(x) = 0$ and $\lim\limits_{x\to\infty} F(x) = 1$.
3. $F(x)$ is right-continuous.

There are three fundamentally different types of probability distributions:

1. **Discrete distributions;**
2. **Singular distributions;**
3. **Absolutely continuous distributions.**

Moreover, every probability distribution can be uniquely decomposed into a mixture of these three types.

A **singular distribution** is concentrated on an uncountable set of Lebesgue measure zero, while assigning probability zero to every individual point.

The most famous example is the **Cantor distribution**, whose cumulative distribution function is known as the *Devil's staircase*.

In this course we will only study **discrete** and **absolutely continuous** distributions.

1 Discrete Probability Distributions

A probability distribution is called **discrete** if there exists a finite or countable set

$$X = \{x_1, x_2, ...\} \subset \mathbb{R}$$

such that

$$P(X) = 1.$$

In other words, the random variable can only take countably many possible values.

If

$$p(x) = P(X = x)$$

denotes the probability mass function (PMF), then the cumulative distribution function is

$$F(y) = \sum_{\substack{x \in X \\ x \leq y}} p(x).$$

Thus, the CDF is obtained by accumulating the probabilities of all values that do not exceed y.

1.1 Example 1: Bernoulli Distribution

Let

$$X \sim \text{Bernoulli}(p).$$

Then

$$P(X = 0) = 1 - p, \qquad P(X = 1) = p.$$

Its cumulative distribution function is

$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - p, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases}$$

Notice that the CDF is a step function with jumps equal to the probabilities of the corresponding outcomes.

1.2 Example 2: Uniform Distribution on a Finite Set

Suppose

$$X = \{x_1, x_2, \dots, x_n\}$$

and every value is equally likely.

Then

$$P(X = x_i) = \frac{1}{n}, \qquad i = 1, \dots, n.$$

Its cumulative distribution function increases by exactly

$$\frac{1}{n}$$

at every point of the support.